

MANDALA: An E(3)-Equivariant Graph Neural Network Framework for Learning Sparse Electronic-Structure Hamiltonian and Density Matrices with Energy and Force Supervision

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Abstract

We present Mandala, a modular software framework for learning block-sparse electronic-structure matrices with E(3)-equivariant graph neural networks. The framework is built around a unified representation of atom-resolved Hamiltonian, overlap, and density matrices, together with reusable abstractions for basis conversion, sparse block handling, irrep mapping, graph construction, model definition, and training. This design allows Mandala to support multiple electronic-structure backends, heterogeneous chemical compositions, and a wide range of neural architecture variants within one workflow.

A distinguishing feature of Mandala is that it does not stop at matrix reconstruction. Predicted operators can be contracted to obtain energy observables and electron counts, differentiated with respect to atomic coordinates to obtain forces, and differentiated with respect to the simulation cell to obtain stress tensors. The predicted Hamiltonian also preserves access to Hamiltonian-based post-processing, including band-structure analysis. This allows the software to bridge electronic-structure learning and observable-guided atomistic modeling while retaining a representation tied to quantum-mechanical operators rather than only scalar targets.

Mandala combines a modular software architecture with an operator-

centered learning formulation for sparse electronic-structure prediction. We emphasize the sparse matrix data model, the observable-guidance loss design, and the broad hyperparameter and architecture search space available in the framework. Numerical results for evaluated systems [TODO: list evaluated systems] are provided in Section [TODO: numerical results section].

Keywords: Electronic structure, Density functional theory, Equivariant graph neural networks, Sparse matrix learning, Energy and force prediction, Scientific machine learning

PROGRAM SUMMARY

Program Title: Mandala

CPC Library link to program files: [TODO: to be added by the Technical Editor]

Developer’s repository link: <https://github.com/Mandala-org/Mandala>

Licensing provisions: Apache-2.0

Programming language: Python

Nature of problem: Electronic-structure calculations based on Kohn-Sham density functional theory remain a central tool in computational materials science and chemistry, but their computational cost limits accessible system sizes, trajectory lengths, and large-scale model-selection studies. Machine-learned interatomic potentials alleviate this cost for energies and forces, yet they generally do not expose the electronic operators needed for richer observables or electronic post-processing. A complementary route is to learn sparse representations of electronic-structure matrices themselves. Such a program must support multiple atom types, changing orbital bases, periodic boundary conditions, equivariant geometric processing, and physically meaningful observable targets such as energies, forces, and stress.

Solution method: Mandala represents Hamiltonian, overlap, and density matrices as atom-pair-resolved sparse block tensors and maps these blocks to irreducible $E(3)$ representations compatible with equivariant neural networks. The package combines parser modules for electronic-structure data, basis-conversion utilities, a shared irrep mapper, graph construction routines, and configurable $E(3)$ -equivariant message-passing models. Multi-head prediction allows simultaneous learning of several matrices, while observable-level quantities are evaluated from the predicted operators using sparse traces. Observable-guidance losses can be added during training, while forces and stress are obtained through automatic differentiation of predicted energies with respect to atomic coordinates and cell degrees of freedom. The framework is organized to support extensibility, architecture search, and reproducible training workflows.

Additional comments including restrictions and unusual features: The framework is optimized for sparse matrix learning in atomistic systems with localized orbital bases and periodic or non-periodic geometries. A distinctive feature of Mandala is the combination of operator-level learning with optional observable guidance derived from those operators, which enables a single software stack to serve both electronic-structure emulation and observable-guided atomistic applications. The package is designed to accommodate multiple data backends and a wide search space of equivariant architectures and training objectives.

1. Introduction

Kohn-Sham density functional theory (DFT) is one of the standard workhorses of electronic-structure simulation because it offers a practical compromise between physical fidelity and computational cost across chemistry, condensed-matter physics, and materials science [1, 2]. Even so, repeated self-consistent solution of the Kohn-Sham equations remains a major bottleneck in workflows that require large supercells, long molecular-dynamics trajectories, broad compositional screening, or repeated re-evaluation under changing thermodynamic conditions. This cost has motivated sustained interest in machine-learned surrogates for quantum-mechanical simulation, including direct learning of densities, wave-function-related quantities, and reduced electronic representations [3–6].

Many successful atomistic machine-learning approaches bypass the explicit electronic structure and instead learn scalar or vector observables such as energies and forces. This strategy is highly effective for interatomic potentials, and recent work has shown that equivariant neural architectures are especially well suited for energy and force prediction because they can encode rotational structure directly in the model rather than relying entirely on handcrafted invariants or data augmentation [7, 8]. However, learning only energies and forces also limits what the surrogate can expose: electronic observables, basis-dependent quantities, operator-level post-processing, and Hamiltonian-derived band-structure information are no longer directly available. For applications that remain fundamentally electronic-structure-centered, it is attractive to learn the operators themselves.

Mandala is designed around this operator-learning viewpoint. The framework targets block-sparse matrix quantities arising in localized-orbital electronic-structure calculations, in particular the Hamiltonian, overlap, and density matrices. Instead of treating these objects as dense global arrays, Mandala

represents them as atom-pair-resolved sparse blocks and combines this representation with $E(3)$ -equivariant graph neural networks. This yields a learning problem that respects both the locality of the underlying electronic structure and the geometric transformation laws imposed by three-dimensional space.

The software is motivated by three specific design goals. First, it should be modular at the level of both scientific abstractions and code organization. Data ingestion, basis conversion, graph construction, irrep mapping, model definition, and training logic should be independently replaceable while still composing into a single workflow. Second, the framework should support supervision beyond raw matrix reconstruction. When Hamiltonian, overlap, and density matrices are predicted jointly, physically relevant observables such as energies and electron counts can be computed directly from the predicted operators, atomic forces can be obtained through differentiation with respect to atomic coordinates, and for periodic systems the same pathway extends naturally to stress tensors. Third, the framework should make architecture exploration a first-class capability rather than an afterthought. Choices such as irreducible feature content, message-passing depth, tensor-product design, head structure, residual connections, normalization, and observable losses must all be accessible through a coherent configuration and sweep interface.

These goals distinguish Mandala from earlier machine-learning workflows in two ways. Relative to purely observable-driven interatomic models, Mandala preserves an operator-level representation that remains useful for electronic-structure analysis. Relative to monolithic research scripts built around one specific dataset or one fixed neural architecture, it provides a reusable framework in which data formats, physical targets, and model variants can be exchanged without rewriting the entire code path. In this sense, Mandala is intended as a unified software environment for sparse operator learning, observable guidance, and architecture exploration.

The following sections describe the sparse matrix data model, the $E(3)$ -equivariant learning formulation, the observable-aware loss construction, the force-training pathway, and the software abstractions that make these components composable. Numerical benchmarks on [TODO: list benchmark systems] and application-specific results for [TODO: list target applications] are reported in Section [TODO: numerical results section].

2. Theoretical Background

2.1. From the full electron-nuclear Schrödinger equation to Kohn-Sham DFT

At the most fundamental nonrelativistic level, an atomistic system is described by the stationary many-body Schrödinger equation for all electrons and all nuclei,

$$\hat{H}(\mathbf{r}; \mathbf{R})\Psi(\mathbf{r}; \mathbf{R}) = E\Psi(\mathbf{r}; \mathbf{R}), \quad (1)$$

where $\Psi(\mathbf{r}; \mathbf{R})$ denotes the many-body wavefunction, $\mathbf{r}$ the collection of electronic coordinates, and $\mathbf{R}$ the collection of ionic coordinates. The semicolon indicates that the dependence on ionic coordinates is parametric within the Born-Oppenheimer approximation. We adopt atomic units throughout, with $\hbar = m_e = e^2 = 1$, so that energies are measured in Hartree and lengths in Bohr radii.

Under the Born-Oppenheimer approximation [9], the electronic Hamiltonian is written as

$$= \hat{T}_e(\mathbf{r}) + \hat{V}_{ee}(\mathbf{r}) + \hat{V}_{ei}(\mathbf{r}; \mathbf{R}) + E_{ii}(\mathbf{R}), \quad (2)$$

with

$$\hat{T}_e(\mathbf{r}) = -\frac{1}{2} \sum_{j=1}^{N_e} \nabla_j^2, \quad (3)$$

$$\hat{V}_{ee}(\mathbf{r}) = \frac{1}{2} \sum_{j=1}^{N_e} \sum_{k \neq j}^{N_e} \frac{1}{|\mathbf{r}_j - \mathbf{r}_k|}, \quad (4)$$

$$\hat{V}_{ei}(\mathbf{r}; \mathbf{R}) = -\sum_{j=1}^{N_e} \sum_{\alpha=1}^{N_i} \frac{Z_\alpha}{|\mathbf{r}_j - \mathbf{R}_\alpha|}, \quad (5)$$

and

$$E_{ii}(\mathbf{R}) = \frac{1}{2} \sum_{\alpha=1}^{N_i} \sum_{\beta \neq \alpha}^{N_i} \frac{Z_\alpha Z_\beta}{|\mathbf{R}_\alpha - \mathbf{R}_\beta|}. \quad (6)$$

The ion-ion term amounts to a constant energy shift for fixed ionic configuration, while the Hamiltonian and wavefunction depend only parametrically on $\mathbf{R}$. In the following, this parametric dependence is not always written explicitly.

The central remaining difficulty is therefore the interacting many-electron problem defined by $\hat{T}_e + \hat{V}_{ee} + \hat{V}_{ei}$ at fixed ionic geometry.

Density functional theory (DFT) replaces the many-electron wavefunction by the electron density as the central variable [1]. In the Kohn-Sham construction [2], the interacting electron problem is mapped onto a fictitious system of non-interacting electrons governed by

$$\left[-\frac{1}{2}\nabla^2 + v_S(\mathbf{r})\right]\psi_j(\mathbf{r}) = \epsilon_j\psi_j(\mathbf{r}), \quad (7)$$

where $j = 1, \dots, N_e$ labels the Kohn-Sham orbitals. The corresponding electronic density is

$$n(\mathbf{r}) = \sum_{j=1}^{N_e} |\psi_j(\mathbf{r})|^2. \quad (8)$$

The Kohn-Sham potential is

$$v_S(\mathbf{r}) = \frac{\delta E_H[n]}{\delta n(\mathbf{r})} + \frac{\delta E_{XC}[n]}{\delta n(\mathbf{r})} + v_{ei}(\mathbf{r}), \quad (9)$$

where $E_H[n]$ is the Hartree energy, $E_{XC}[n]$ the exchange-correlation energy, and

$$v_{ei}(\mathbf{r}) = -\sum_{\alpha} \frac{Z_{\alpha}}{|\mathbf{r} - \mathbf{R}_{\alpha}|}. \quad (10)$$

Although the Kohn-Sham construction is formally exact, practical calculations rely on approximations to E_{XC} .

In this notation, the ground-state energy functional is written as

$$E[n] = T_S[n] + E_H[n] + E_{XC}[n] + \int d\mathbf{r} n(\mathbf{r})v_{ei}(\mathbf{r}) + E_{ii}, \quad (11)$$

where $T_S[n]$ is the Kohn-Sham kinetic energy. This is the DFT-level starting point that Mandala connects to localized-orbital operator learning.

Mandala operates on localized-orbital matrix representations of this problem. Expanding the Kohn-Sham orbitals in a finite basis $\{\chi_{\mu}\}$,

$$\psi_j(\mathbf{r}) = \sum_{\mu} C_{\mu j} \chi_{\mu}(\mathbf{r}), \quad (12)$$

leads to the generalized eigenvalue problem

$$\sum_{\nu} H_{\mu\nu} C_{\nu j} = \epsilon_j \sum_{\nu} S_{\mu\nu} C_{\nu j}, \quad (13)$$

with

$$H_{\mu\nu} = \langle \chi_\mu | \hat{H}_{\text{KS}} | \chi_\nu \rangle, \quad S_{\mu\nu} = \langle \chi_\mu | \chi_\nu \rangle. \quad (14)$$

The one-particle density matrix is

$$D_{\mu\nu} = \sum_j f_j C_{\mu j} C_{\nu j}^*, \quad (15)$$

where f_j denotes the orbital occupation. In a nonorthogonal basis, the electron count is

$$N_e = \text{Tr}(DS), \quad (16)$$

while the energy observable used throughout Mandala is obtained from

$$E_{\text{op}} = \text{Tr}(DH). \quad (17)$$

For backend-specific total energies, additional double-counting and ion-ion terms may be required, but the trace expressions above are the core operator identities used throughout Mandala.

Localized basis sets also induce block sparsity. If orbital μ belongs to atom i and orbital ν belongs to atom j , then matrix entries can be organized into atom-pair blocks $X_{ij}^{\mathbf{L}}$, where $X \in \{H, S, D\}$ and $\mathbf{L} \in \mathbb{Z}^3$ indexes periodic lattice images. This sparse block structure is the operator domain learned by Mandala.

2.1.1. Hamiltonian gauge invariance and the overlap-shift correction

In a nonorthogonal basis, the generalized eigenproblem is invariant under a rigid shift of the Hamiltonian by a multiple of the overlap matrix. Let

$$HC = SC\varepsilon, \quad (18)$$

and define a shifted Hamiltonian

$$\tilde{H} = H - \alpha S. \quad (19)$$

Then

$$\tilde{H}C = (H - \alpha S)C = SC\varepsilon - \alpha SC = SC(\varepsilon - \alpha I). \quad (20)$$

Therefore the generalized eigenvectors are unchanged, while all eigenvalues are shifted by the same constant α . This proves that the choice of zero of energy can be represented, in matrix form, as the transformation $H \mapsto$

$H - \alpha S$. For a fixed electron number and a geometry-independent shift α , one also finds

$$\text{Tr}(D\tilde{H}) = \text{Tr}(DH) - \alpha\text{Tr}(DS) = \text{Tr}(DH) - \alpha N_e, \quad (21)$$

so only the absolute energy reference changes.

Mandala exploits this gauge freedom through the overlap-shift correction μ_H , which is chosen to minimize the mean-squared discrepancy between a predicted Hamiltonian and a reference Hamiltonian after subtraction of a multiple of the overlap matrix:

$$\mu_H = \arg \min_{\mu} \|H^{\text{pred}} - H^{\text{ref}} - \mu S\|_F^2. \quad (22)$$

Taking the derivative with respect to μ and setting it to zero yields

$$\mu_H = \frac{\langle H^{\text{pred}} - H^{\text{ref}}, S \rangle_F}{\langle S, S \rangle_F}, \quad (23)$$

where $\langle A, B \rangle_F = \sum_{\mu\nu} A_{\mu\nu} B_{\mu\nu}$ is the Frobenius inner product, evaluated in Mandala blockwise over the aligned sparse representation. The corrected Hamiltonian $H^{\text{pred}} - \mu_H S$ therefore removes a pure gauge offset without altering the physically relevant eigenvectors.

2.2. $E(3)$ symmetry, equivariance, and irreducible representations

Atomistic systems in three-dimensional space are naturally acted on by the Euclidean group $E(3) = \mathbb{R}^3 \rtimes O(3)$, combining translations, rotations, and reflections. For a group element $g = (Q, \mathbf{t})$, with $Q \in O(3)$ and $\mathbf{t} \in \mathbb{R}^3$, atomic positions transform as

$$\mathbf{R}_i \mapsto g \cdot \mathbf{R}_i = Q\mathbf{R}_i + \mathbf{t}. \quad (24)$$

A model output $f(X)$ is invariant if $f(g \cdot X) = f(X)$, and equivariant if

$$f(g \cdot X) = \rho_{\text{out}}(g) f(X), \quad (25)$$

for a representation ρ_{out} appropriate to the predicted object. Scalar observables such as total energies are invariant. Vectors such as forces transform covariantly, and operator-valued quantities inherit structured transformation rules from the orbital basis in which they are expressed.

In equivariant neural networks, hidden features are not treated as generic channels. Instead, they are decomposed into irreducible representations (irreps) of $O(3)$, labeled by angular momentum ℓ and parity $p \in \{+1, -1\}$. A feature space therefore takes the form

$$\mathcal{H} = \bigoplus_{\ell,p} m_{\ell,p} D^{(\ell,p)}, \quad (26)$$

where $m_{\ell,p}$ is the multiplicity of irrep (ℓ, p) and $D^{(\ell,p)}$ is the corresponding representation matrix. Scalars correspond to $(\ell = 0, p = +1)$, ordinary vectors to $(\ell = 1, p = -1)$, and higher-rank objects are built from larger ℓ channels. This representation-theoretic bookkeeping is particularly natural for localized atomic orbitals, whose angular content is already organized by orbital angular momentum.

The key architectural consequence is that every learned operation must respect the decomposition into irreps. Node and edge features are updated through maps that commute with the group action, rather than by unrestricted dense linear layers. Geometric information enters through relative displacement vectors and their spherical-harmonic expansions, while translational invariance is enforced by expressing all geometry in relative coordinates. Reflections are tracked through the parity label p , while rotations act within each ℓ channel. In this way, Mandala preserves directional information throughout the network without sacrificing the required covariance laws [7, 8].

2.3. Tensor products, spherical harmonics, and the Wigner-Eckart viewpoint

The basic algebraic operation of an equivariant graph neural network is the tensor product of irreps. If two feature channels transform according to $D^{(\ell_1,p_1)}$ and $D^{(\ell_2,p_2)}$, then their tensor product decomposes as

$$D^{(\ell_1,p_1)} \otimes D^{(\ell_2,p_2)} = \bigoplus_{L=|\ell_1-\ell_2|}^{\ell_1+\ell_2} D^{(L,p_1p_2)}. \quad (27)$$

This Clebsch-Gordan structure determines which output irreps may be produced by combining two input channels. In practical terms, it is the algebraic rule that governs message passing, nonlinear mixing of hidden features, and the final projection from latent features to matrix coefficients.

Edge geometry is injected through spherical harmonics $Y_{\ell m}(\hat{\mathbf{r}}_{ij})$, evaluated on normalized relative displacement directions. Because spherical harmonics transform irreducibly under rotations, they provide the canonical angular basis for equivariant edge messages. A typical equivariant interaction therefore combines learned node or edge features with spherical-harmonic features through tensor products, while radial functions modulate the interaction strength as a function of distance.

The same representation-theoretic structure appears in matrix elements of operators between angular-momentum-carrying orbitals. For a spherical tensor operator $T_q^{(k)}$, the Wigner-Eckart theorem states that

$$\langle \alpha \ell m | T_q^{(k)} | \alpha' \ell' m' \rangle = \langle \ell' m'; k q | \ell m \rangle \langle \alpha \ell || T^{(k)} || \alpha' \ell' \rangle, \quad (28)$$

that is, the dependence on magnetic quantum numbers is separated from the reduced matrix element. Mandala uses this viewpoint operationally: the angular coupling structure is fixed analytically by the irrep mapping, while the neural network learns the reduced coefficients associated with each orbital pair and chemical environment. This is precisely why the matrix-block prediction problem can be posed naturally in an irrep basis rather than directly in Cartesian matrix entries.

2.4. From equivariant local features to sparse operator prediction

Mandala represents each operator $X \in \{H, S, D\}$ as a sparse family of atom-pair blocks $X_{ij}^{\mathbf{L}}$, where i and j index atoms and $\mathbf{L}$ labels periodic images. The block representation is converted into an irrep representation by a fixed change of basis,

$$\mathbf{x}_{ij}^{\mathbf{L}} = Q_{a_i a_j} \text{vec}(X_{ij}^{\mathbf{L}}), \quad (29)$$

where a_i and a_j denote the atomic species of atoms i and j , and $Q_{a_i a_j}$ is the species-pair-specific block-to-irrep transform. The inverse map reconstructs matrix blocks from predicted irrep vectors,

$$\text{vec}(X_{ij}^{\mathbf{L}}) = Q_{a_i a_j}^{-1} \mathbf{x}_{ij}^{\mathbf{L}}, \quad (30)$$

which, in the orthonormal convention used internally, is implemented as a transpose. In the code, these maps are materialized by the `BlockIrrepMapper`, which is reused by the data pipeline, the model heads, and the evaluation routines.

Once the network predicts a consistent set of sparse operators, observable quantities are obtained directly from those operators rather than through a separate surrogate model. For sparse block matrices, the trace of a product is evaluated as

$$\text{Tr}(AB) = \sum_{(\mathbf{L}, i, j)} \text{tr}(A_{ij}^{\mathbf{L}} B_{ji}^{-\mathbf{L}}), \quad (31)$$

which is exactly the sparse contraction implemented by Mandala. This gives direct access to electron counts, energy observables, and derivative observables such as forces and stress, all from the same predicted operator set. The resulting workflow differs from direct scalar regression: internal consistency between matrices and observables is built into the model formulation rather than imposed only at the level of post hoc analysis.

3. Numerical Implementation

3.1. Software design goals, sparse data model, and periodic graph construction

Mandala is organized as a layered scientific software stack whose core workflow consists of parsing electronic-structure outputs, harmonizing basis conventions, building sparse operators and graphs, predicting irrep vectors, reconstructing operators, and evaluating observables. The central data abstractions are `BlockMatrix`, `IrrepsBlockData`, and `Snapshot`. A `Snapshot` contains all matrices and metadata associated with one geometry, including positions, cell, optional forces, and optional stress targets. This keeps Hamiltonian, overlap, and density predictions tied to a single physically consistent basis and sparse indexing scheme.

The sparse operator representation stores blocks by ordered species pair and by explicit edge metadata $(\mathbf{L}, i, j)$, where $\mathbf{L} \in \mathbb{Z}^3$ is the lattice-image shift. Basis-conversion modules map backend-specific orbital orderings and sign conventions into the common internal representation expected by the equivariant stack. In the current framework, OpenMX and FHI-aims are fully supported, and PySCF data are handled through the same sparse operator interface [TODO: finish PySCF training-data integration].

Graph construction mirrors the sparse operator indexing. Nodes correspond to atoms, while edges correspond to self-interactions and neighbor interactions inside a cutoff radius. For periodic systems, each edge carries an

integer shift vector $\mathbf{L}$ and a relative displacement

$$\mathbf{r}_{ij}^L = \mathbf{R}_j - \mathbf{R}_i + \sum_{\alpha=1}^3 L_{\alpha} \mathbf{a}_{\alpha}, \quad (32)$$

where $\{\mathbf{a}_{\alpha}\}$ are the cell vectors. These displacements are used to construct radial embeddings and spherical-harmonic features. To preserve exact agreement between graph edges and sparse matrix blocks, Mandala imposes a canonical ordering of edges by distance and lattice shift, with reverse-edge lookup tables used for sparse traces and symmetrization. This explicit reverse alignment is essential for both physical correctness and reproducibility under periodic boundary conditions.

The same snapshot-level representation preserves access to operator-derived postprocessing. In particular, the predicted Hamiltonian may be reused for band-structure analysis without leaving the sparse operator workflow [TODO: specify workflow/API entry point].

3.2. $E(3)$ -equivariant message passing, readout heads, and architecture variants

The network is composed of node encoders, edge encoders, a stack of equivariant message-passing blocks, and matrix-specific readout heads. Node encoders embed chemical species into scalar channels. Edge encoders combine edge-type embeddings, radial distance embeddings, and spherical harmonics into equivariant edge features. If $\mathbf{h}_i^{(t)}$ denotes node features and $\mathbf{e}_{ij}^{(t)}$ denotes edge features at layer t , a generic Mandala message-passing step may be written as

$$\mathbf{m}_{ij}^{(t)} = \Phi_{\text{edge}}^{(t)} \left(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}, \mathbf{e}_{ij}^{(t)}, Y(\hat{\mathbf{r}}_{ij}), \varphi(r_{ij}) \right), \quad (33)$$

$$\mathbf{h}_i^{(t+1)} = \Phi_{\text{node}}^{(t)} \left(\mathbf{h}_i^{(t)}, \text{Agg}_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}^{(t)} \right), \quad (34)$$

where $\Phi_{\text{edge}}^{(t)}$ and $\Phi_{\text{node}}^{(t)}$ are equivariant maps assembled from tensor products, equivariant linear layers, radial MLPs, parity-aware nonlinearities, and optional residual connections, while Agg denotes a configurable aggregation rule such as summation or attention.

The readout stage maps final latent features to the irrep content required by each matrix target. For a target operator X , the head predicts irrep vectors

$$\hat{\mathbf{x}}_{ij}^{L,X} = \Psi_X \left(\mathbf{e}_{ij}^{(L)}, \mathbf{h}_i^{(L)}, \mathbf{h}_j^{(L)}, \varphi(r_{ij}) \right), \quad (35)$$

followed by inverse irrep-to-block conversion

$$\hat{X}_{ij}^L = Q_{a_i a_j}^\top \hat{\mathbf{x}}_{ij}^{L,X}. \quad (36)$$

Because heads share the same equivariant latent representation, Mandala can predict Hamiltonian, overlap, and density matrices jointly with only modest additional readout cost.

An important feature of the framework is that the architecture is not fixed to one canonical model. Mandala exposes a broad family of variants within the same software interface. Search dimensions include the hidden irrep content, the angular cutoff $\ell_{\max}$, the number of message-passing layers, the radial basis size, the depth of the neck and head MLPs, and the choice of tensor product implementation. The message-passing blocks support alternative edge updates (tensor-product, concatenation-based, and replacement-style variants), alternative node updates (sum, concatenation, and replacement variants), alternative aggregation mechanisms (including attention-style aggregation), pre- and post-linear transformations, optional self-connections, and optional node and edge residual paths.

The encoder and head stages are likewise configurable. Edge encoders support various variants, together with optional tensor-square feature expansion of spherical-harmonics. Readout heads support direct equivariant projections, deeper equivariant MLP heads, optional tensor-square preconditioning of hidden features, scalar-specific auxiliary branches, optional log-scale or magnitude factorization for output amplitudes, optional use of node embeddings for self-edges, and separate treatment of diagonal, shifted-self, and off-diagonal blocks. Because all of these variants share the same block-sparse data model and irrep mapping layer, architecture search can be performed without changing the surrounding training and evaluation workflow.

3.3. Sparse operator algebra, symmetrization, gauge correction, and density normalization

Sparse operator manipulations are implemented directly on aligned block data. Given two sparse operators A and B with reverse-edge alignment already established, Mandala evaluates their trace contraction as

$$\text{Tr}(AB) = \sum_{(L,i,j)} \text{tr}(A_{ij}^L B_{ji}^{-L}), \quad (37)$$

without assembling dense global matrices. This operator algebra is reused for energy and electron-count evaluation, loss construction, and diagnostic analysis.

For matrix targets expected to be symmetric or Hermitian in the chosen basis, Mandala supports explicit sparse symmetrization after block reconstruction. In the real-valued convention used by the implementation, this takes the form

$$X_{ij}^L \leftarrow \frac{1}{2} \left(X_{ij}^L + (X_{ji}^L)^\top \right). \quad (38)$$

This operation is carried out using precomputed reverse-edge permutations, so that symmetry is enforced consistently across periodic images and self-edges.

Hamiltonian gauge freedom is handled operationally through the overlap-shift correction described in Section 2.1.1. In practical evaluation and analysis workflows, Mandala computes

$$\tilde{H}^{\text{pred}} = H^{\text{pred}} - \mu_H S, \quad \mu_H = \frac{\langle H^{\text{pred}} - H^{\text{ref}}, S \rangle_F}{\langle S, S \rangle_F}, \quad (39)$$

so that purely gauge-like Hamiltonian offsets are removed before reporting gauge-corrected operator errors.

The framework also supports density normalization to the correct number of electrons. If a predicted density $\hat{D}$ and overlap $\hat{S}$ produce $\hat{N}_e = \text{Tr}(\hat{D}\hat{S})$, then a normalized density is obtained by the scalar rescaling

$$\hat{D}_{\text{norm}} = \beta \hat{D}, \quad \beta = \frac{N_e^{\text{target}}}{\text{Tr}(\hat{D}\hat{S})}. \quad (40)$$

This provides a simple and effective way to enforce electron-count consistency.

Together, sparse trace evaluation, symmetry restoration, gauge correction, and density normalization define a compact postprocessing layer that turns raw network outputs into physically interpretable operators ready for observable evaluation and Hamiltonian-based analyses such as band-structure calculation.

3.4. Observable guidance with energies, electron counts, forces, and stress

Mandala supports pure matrix learning, mixed matrix-and-observable learning, and fully observable-guided training from one common operator-prediction pipeline. For a chosen set of matrix targets $\mathcal{M} \subseteq \{H, S, D\}$, the

general training objective is

$$\mathcal{L} = \sum_{X \in \mathcal{M}} \lambda_X \mathcal{L}_X^{\text{matrix}} + \lambda_E \mathcal{L}^E + \lambda_N \mathcal{L}^N + \lambda_F \mathcal{L}^F + \lambda_\sigma \mathcal{L}^\sigma. \quad (41)$$

The matrix terms compare predicted and reference sparse operators, typically by a weighted combination of ℓ_1 and ℓ_2 errors,

$$\mathcal{L}_X^{\text{matrix}} = \eta \text{MAE}(\hat{X}, X) + (1 - \eta) \text{MSE}(\hat{X}, X), \quad (42)$$

with η controlled by configuration.

Observable guidance is defined on quantities derived from the predicted operators. In the operator-centered workflow used throughout Mandala, the energy observable used in training and the number of electrons are

$$\hat{E} = \text{Tr}(\hat{D}\hat{H}), \quad \hat{N}_e = \text{Tr}(\hat{D}\hat{S}), \quad (43)$$

which gives

$$\mathcal{L}^E = \text{MSE}(\hat{E}, E^{\text{ref}}), \quad \mathcal{L}^N = \text{MSE}(\hat{N}_e, N_e^{\text{ref}}). \quad (44)$$

The framework also supports mixed observable guidance paths in which one of the operators entering the trace is replaced by the reference operator, for example $\text{Tr}(\hat{H}D^{\text{ref}})$, $\text{Tr}(\hat{D}H^{\text{ref}})$, $\text{Tr}(\hat{D}S^{\text{ref}})$, or $\text{Tr}(D^{\text{ref}}\hat{S})$, enabling diagnostic and training modes that separate errors due to different predicted operators.

Forces are obtained by automatic differentiation of the predicted scalar energy with respect to atomic coordinates,

$$\hat{\mathbf{F}}_I = -\frac{\partial \hat{E}}{\partial \mathbf{R}_I}, \quad (45)$$

and the corresponding force loss is

$$\mathcal{L}^F = \frac{1}{3N_{\text{atoms}}} \sum_I \left\| \hat{\mathbf{F}}_I - \mathbf{F}_I^{\text{ref}} \right\|_2^2. \quad (46)$$

No separate force head is required: forces are constrained through the same predicted operators that define the energy.

For periodic systems, stress guidance is obtained by differentiating the predicted energy with respect to the cell matrix h . If $\Omega = \det(h)$ is the cell volume, Mandala uses

$$\hat{\sigma} = \frac{1}{\Omega} h^\top \frac{\partial \hat{E}}{\partial h}, \quad (47)$$

with optional symmetrization

$$\hat{\sigma} \leftarrow \frac{1}{2} (\hat{\sigma} + \hat{\sigma}^\top), \quad (48)$$

and defines the stress loss as

$$\mathcal{L}^\sigma = \frac{1}{9} \|\hat{\sigma} - \sigma^{\text{ref}}\|_F^2. \quad (49)$$

This completes the observable-guidance picture: one can train a single operator-valued model under simultaneous constraints from matrix entries, energies, electron counts, forces, and stress tensors.

3.5. Configuration-driven architecture search

Because the same sparse operator workflow supports many architectural choices, Mandala treats model search as a primary scientific use case rather than as an auxiliary engineering convenience. All major representation, encoder, message-passing, readout, and loss-weighting choices are exposed as structured configuration parameters and can therefore be swept systematically.

Search dimensions include hidden irreps, angular cutoff, radial basis size, message-passing depth, encoder style, tensor-product type, normalization and nonlinearity families, node and edge update variants, residual pathways, neck and head depth, separate treatment of shifted-self blocks, tensor-square readout options, scalar-specific branches, observable-loss coefficients, force and stress weights, and the subset of matrix targets predicted jointly. This wide configuration surface is one of the main scientific strengths of the framework: it enables controlled comparisons between architecture families without forcing changes to the data model, parser layer, or observable-evaluation stack.

3.6. Verification, reproducibility, and extensibility

Mandala includes tests and validation routines covering data parsing, basis harmonization, sparse operator alignment, block-to-irrep mapping, graph construction, equivariant tensor-product paths, observable evaluation, and end-to-end training workflows. These checks are especially important in a code base where subtle mistakes in basis ordering, reverse-edge alignment, or irrep bookkeeping can produce apparently plausible but physically inconsistent results.

Reproducibility follows from configuration-driven runs, explicit snapshot serialization, deterministic sparse edge ordering, and checkpointed training. Extensibility follows from the layered structure described above: new electronic structure backends, new observable definitions, new equivariant blocks, and new training objectives can be incorporated without changing the core operator data model. This design is what allows Mandala to function both as a reusable scientific software package and as a framework for rapid methodological experimentation.

4. Conclusion

Mandala is a scientific software framework for learning sparse electronic-structure operators with $E(3)$ -equivariant graph neural networks. Its defining features are the explicit sparse block representation of Hamiltonian, overlap, and density matrices; the modular software layers that connect parsing, basis harmonization, irrep mapping, graph construction, model definition, and training; and the ability to supervise not only matrices themselves but also observables derived from them.

The framework is designed to connect two communities that are often separated in practice. From the electronic-structure side, it retains access to operator-level information and basis-aware quantities. From the atomistic machine-learning side, it inherits the flexibility of modern equivariant message passing, multitask learning, and automated hyperparameter search. By combining these perspectives, Mandala supports workflows in which one can train on matrix fidelity, energy-observable fidelity, electron-count fidelity, force fidelity, and stress fidelity within one coherent implementation.

Quantitative benchmarks on [TODO: list systems], scientific case studies for [TODO: list applications], and comparative assessments against [TODO: list baseline methods] are reported in Section [TODO: numerical results section]. Mandala is therefore positioned not merely as a prototype model, but as a modular platform for operator-level machine learning in electronic-structure simulation.

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